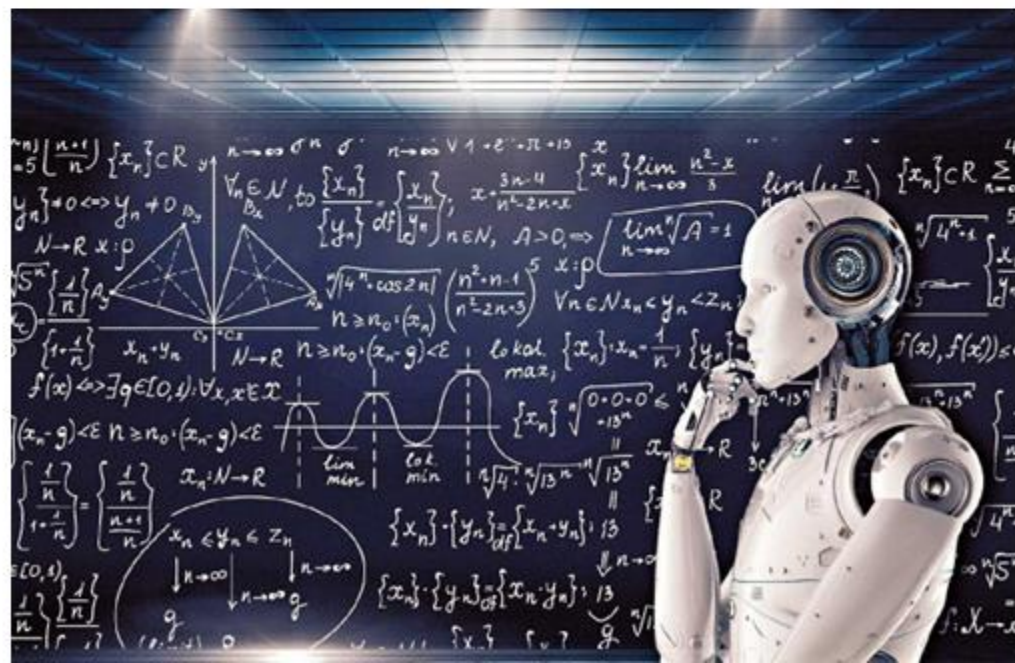


Linear Discriminant Analysis



Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction and classification technique. Unlike PCA, which ignores class labels, LDA uses class information to find feature combinations that best separate different classes.

LDA is widely used in:

- Pattern recognition
- Face recognition
- Medical diagnosis
- Text classification

Intuition of LDA

Suppose we have two classes of data points.

Goal of LDA:

- Maximize separation between classes
- Minimize variance within each class

Thus LDA tries to find a projection direction where:

- Class means are far apart
- Points within each class are close together

This produces maximum class separability.

PCA vs LDA

Feature

PCA

LDA

Type

Unsupervised

Supervised

Uses class labels

No

Yes

Objective

Maximize variance

Maximize class separation

Method

Eigenvectors of
covariance matrix

Eigenvectors of scatter
matrices

Key Idea of LDA

LDA finds a projection vector w that maximizes the Fisher criterion:

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

Where:

- S_B = between-class scatter matrix
- S_W = within-class scatter matrix

Goal:

maximize $J(w)$

Scatter Matrices

Within-Class Scatter Matrix

Measures how spread out samples are **inside each class**.

$$S_W = \sum_{k=1}^c \sum_{x_i \in C_k} (x_i - \mu_k)(x_i - \mu_k)^T$$

Where:

- c = number of classes
- x_i = data point
- μ_k = mean of class k

Between-Class Scatter Matrix

Measures separation between class means.

$$S_B = \sum_{k=1}^c n_k (\mu_k - \mu)(\mu_k - \mu)^T$$

Where:

- n_k = samples in class k
- μ_k = class mean
- μ = overall mean

Optimal Projection

LDA solves

$$S_W^{-1} S_B w = \lambda w$$

Thus we compute:

- eigenvalues
- eigenvectors

The eigenvector with the largest eigenvalue gives the best discriminant direction.

Mathematical Example

- Suppose we have two classes with two features.

Class 1

x1	x2
2	3
3	3
2	2

Class 2

x1	x2
7	8
8	8
7	7

Mathematical Example

Step 1 — Compute Class Means

Class 1 mean:

$$\mu_1 = \begin{bmatrix} 2.33 \\ 2.67 \end{bmatrix}$$

Class 2 mean:

$$\mu_2 = \begin{bmatrix} 7.33 \\ 7.67 \end{bmatrix}$$

$$S_W = \sum_{k=1}^c \sum_{x_i \in C_k} (x_i - \mu_k)(x_i - \mu_k)^T$$

Where:

- c = number of classes
- x_i = data point
- μ_k = mean of class k

Step 2 — Compute Within-Class Scatter

For each sample:

$$(x_i - \mu_k)(x_i - \mu_k)^T$$

Example for first sample (2,3):

$$x - \mu_1 = \begin{bmatrix} 2 - 2.33 \\ 3 - 2.67 \end{bmatrix} = \begin{bmatrix} -0.33 \\ 0.33 \end{bmatrix}$$

Outer product:

$$\begin{bmatrix} -0.33 \\ 0.33 \end{bmatrix} \begin{bmatrix} -0.33 & 0.33 \end{bmatrix} \\ = \begin{bmatrix} 0.11 & -0.11 \\ -0.11 & 0.11 \end{bmatrix}$$

Do this for all points and sum.

Result:

$$S_1 = \begin{bmatrix} 0.67 & 0.33 \\ 0.33 & 0.67 \end{bmatrix}$$

Similarly for Class 2:

$$S_2 = \begin{bmatrix} 0.67 & 0.33 \\ 0.33 & 0.67 \end{bmatrix}$$

Thus

$$S_W = S_1 + S_2 \\ S_W = \begin{bmatrix} 1.34 & 0.66 \\ 0.66 & 1.34 \end{bmatrix}$$

Mathematical Example

Step 3 — Between-Class Scatter Matrix Formula

The between-class scatter matrix is

$$S_B = \sum_{k=1}^c n_k (\mu_k - \mu)(\mu_k - \mu)^T$$

Where

- n_k = number of samples in class k
- μ_k = class mean
- μ = overall mean

Since we have two classes, the formula becomes

$$S_B = n_1(\mu_1 - \mu)(\mu_1 - \mu)^T + n_2(\mu_2 - \mu)(\mu_2 - \mu)^T$$

Step 4 — Compute Mean Difference (Class 1)

$$\begin{aligned} & \mu_1 - \mu \\ &= \begin{bmatrix} 2.33 - 4.83 \\ 2.67 - 5.17 \end{bmatrix} \\ &= \begin{bmatrix} -2.5 \\ -2.5 \end{bmatrix} \end{aligned}$$

Step 5 — Compute Outer Product (Class 1)

$$\begin{aligned} & (\mu_1 - \mu)(\mu_1 - \mu)^T \\ &= \begin{bmatrix} -2.5 \\ -2.5 \end{bmatrix} \begin{bmatrix} -2.5 & -2.5 \end{bmatrix} \end{aligned}$$

Multiply:

$$\begin{aligned} &= \begin{bmatrix} (-2.5)(-2.5) & (-2.5)(-2.5) \\ (-2.5)(-2.5) & (-2.5)(-2.5) \end{bmatrix} \\ &= \begin{bmatrix} 6.25 & 6.25 \\ 6.25 & 6.25 \end{bmatrix} \end{aligned}$$

Step 6 — Multiply by Number of Samples

$$\begin{aligned} & n_1 = 3 \\ & 3 \times \begin{bmatrix} 6.25 & 6.25 \\ 6.25 & 6.25 \end{bmatrix} = \begin{bmatrix} 18.75 & 18.75 \\ 18.75 & 18.75 \end{bmatrix} \end{aligned}$$

Mathematical Example

Step 7 — Compute Mean Difference (Class 2)

$$\begin{aligned}\mu_2 - \mu \\&= \begin{bmatrix} 7.33 - 4.83 \\ 7.67 - 5.17 \end{bmatrix} \\&= \begin{bmatrix} 2.5 \\ 2.5 \end{bmatrix}\end{aligned}$$

Step 8 — Compute Outer Product (Class 2)

$$\begin{aligned}(\mu_2 - \mu)(\mu_2 - \mu)^T \\&= \begin{bmatrix} 2.5 \\ 2.5 \end{bmatrix} \begin{bmatrix} 2.5 & 2.5 \end{bmatrix} \\&= \begin{bmatrix} 6.25 & 6.25 \\ 6.25 & 6.25 \end{bmatrix}\end{aligned}$$

Step 9 — Multiply by Number of Samples

$$\begin{aligned}n_2 = 3 \\3 \times \begin{bmatrix} 6.25 & 6.25 \\ 6.25 & 6.25 \end{bmatrix} &= \begin{bmatrix} 18.75 & 18.75 \\ 18.75 & 18.75 \end{bmatrix}\end{aligned}$$

Step 10 — Compute Final Between-Class Scatter Matrix

$$\begin{aligned}S_B &= \begin{bmatrix} 18.75 & 18.75 \\ 18.75 & 18.75 \end{bmatrix} + \begin{bmatrix} 18.75 & 18.75 \\ 18.75 & 18.75 \end{bmatrix} \\&= \begin{bmatrix} 37.5 & 37.5 \\ 37.5 & 37.5 \end{bmatrix}\end{aligned}$$

Compute the Optimal Projection Direction

The main objective of LDA is to find a projection vector w that **maximizes class separability**. This is achieved by solving the generalized eigenvalue problem:

$$S_W^{-1} S_B w = \lambda w$$

where:

- S_W = within-class scatter matrix
- S_B = between-class scatter matrix
- w = projection vector (discriminant direction)
- λ = eigenvalue

The matrix $S_W^{-1} S_B$ captures the relationship between **class separation and within-class variance**.

Eigenvector Represents the Best Discriminant Direction

After computing $S_W^{-1}S_B$, we perform eigenvalue decomposition to obtain eigenvalues and eigenvectors.

The eigenvector corresponding to the largest eigenvalue gives the optimal projection direction.

Mathematical details are as on upcoming slides.

Compute the Optimal Projection Direction

Step 11 — Recall the Matrices

From the previous calculations:

Within-class scatter matrix

$$S_W = \begin{bmatrix} 1.34 & 0.66 \\ 0.66 & 1.34 \end{bmatrix}$$

Between-class scatter matrix

$$S_B = \begin{bmatrix} 37.5 & 37.5 \\ 37.5 & 37.5 \end{bmatrix}$$

Step 12 — Compute the Inverse of S_W

For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Determinant

$$\begin{aligned} |S_W| &= (1.34)(1.34) - (0.66)(0.66) \\ &= 1.7956 - 0.4356 \\ &= 1.36 \end{aligned}$$

Inverse Matrix

$$\begin{aligned} S_W^{-1} &= \frac{1}{1.36} \begin{bmatrix} 1.34 & -0.66 \\ -0.66 & 1.34 \end{bmatrix} \\ &= \begin{bmatrix} 0.985 & -0.485 \\ -0.485 & 0.985 \end{bmatrix} \end{aligned}$$

Compute the Optimal Projection Direction

Step 13 — Compute $S_W^{-1}S_B$

Multiply

$$S_W^{-1} \begin{bmatrix} 37.5 & 37.5 \\ 37.5 & 37.5 \end{bmatrix}$$

First row:

$$\begin{aligned} (0.985)(37.5) + (-0.485)(37.5) \\ = 36.94 - 18.19 \\ = 18.75 \end{aligned}$$

Both columns are identical, so:

$$= \begin{bmatrix} 18.75 & 18.75 \end{bmatrix}$$

Second row:

$$\begin{aligned} (-0.485)(37.5) + (0.985)(37.5) \\ = -18.19 + 36.94 \\ = 18.75 \end{aligned}$$

Thus

$$S_W^{-1}S_B = \begin{bmatrix} 18.75 & 18.75 \\ 18.75 & 18.75 \end{bmatrix}$$

Step 14 — Solve Eigenvalue Equation

LDA solves

$$(S_W^{-1}S_B)w = \lambda w$$

Let

$$A = \begin{bmatrix} 18.75 & 18.75 \\ 18.75 & 18.75 \end{bmatrix}$$

Step 15 — Compute Eigenvalues

Solve

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 18.75 - \lambda & 18.75 \\ 18.75 & 18.75 - \lambda \end{vmatrix} = 0$$

Expand determinant:

$$(18.75 - \lambda)^2 - (18.75)^2 = 0$$

$$\lambda_1 = 37.5$$

$$\lambda_2 = 0$$

The largest eigenvalue is 37.5, so we use its eigenvector.

Compute the Optimal Projection Direction

Step 16 — Compute Eigenvector

Solve

$$(A - \lambda_1 I)w = 0$$
$$A - 37.5I = \begin{bmatrix} -18.75 & 18.75 \\ 18.75 & -18.75 \end{bmatrix}$$

Let

$$w = \begin{bmatrix} x \\ y \end{bmatrix}$$

Multiply:

$$-18.75x + 18.75y = 0$$

Divide by 18.75:

$$-x + y = 0$$
$$y = x$$

Thus eigenvector direction:

$$w = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 17 — Normalize the Eigenvector

Length:

$$\|w\| = \sqrt{1^2 + 1^2}$$
$$= \sqrt{2}$$

Normalized vector:

$$w = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

Approximation:

$$w \approx \begin{bmatrix} 0.71 \\ 0.71 \end{bmatrix}$$

Final Result

The optimal LDA projection direction is

$$w = \begin{bmatrix} 0.71 \\ 0.71 \end{bmatrix}$$

which corresponds to the discriminant function

$$z = x_1 + x_2$$

This direction maximizes the distance between class means while minimizing within-class variance.

Explanation of Result

1. What LDA Is Trying to Do

The goal of **Linear Discriminant Analysis (LDA)** is to find a direction w where:

- **Classes are far apart** (large inter-class variance)
- **Points inside each class are tightly grouped** (small intra-class variance)

In other words, we want **maximum separation between classes but minimum spread inside classes**.

2. Fisher Criterion (Core Idea of LDA)

LDA uses the **Fisher criterion**:

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

Where:

- S_B = between-class scatter (distance between class means)
- S_W = within-class scatter (spread inside each class)
- w = projection direction

The objective is

$$\text{maximize } J(w)$$

Explanation of Result

3. What Our Final Result Means

We found

$$w = \begin{bmatrix} 0.71 \\ 0.71 \end{bmatrix}$$

Since the two components are equal, the direction is proportional to

$$w = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Thus the projection formula becomes

$$z = w^T x$$

$$z = x_1 + x_2$$

This means LDA projects the data onto the line where the sum of the features is measured.

4. Why This Direction Separates the Classes Best

Look at the dataset:

Class 1

x1	x2
2	3
3	3
2	2

Values are around (2–3).

Class 2

x1	x2
7	8
8	8
7	7

Values are around (7–8).

So both variables increase **together** when we move from Class 1 to Class 2.

Graphically, the data moves roughly along the **diagonal direction**.

That diagonal direction is exactly:

Explanation of Result

5. What Happens When We Project onto $x_1 + x_2$

Compute projections:

Class 1

Point	$z = x_1 + x_2$
(2,3)	5
(3,3)	6
(2,2)	4

Range: 4–6

Class 2

Point	$z = x_1 + x_2$
(7,8)	15
(8,8)	16
(7,7)	14

Range: 14–16

Now the classes become **very far apart** on a single axis:

Class 1	Class 2
4 5 6	14 15 16

There is a **large gap** between the classes.

Thank You